

Farm Shed Stand

Outdoor garden storage structure with double doors, shelving system & metal roof

WIDTH 96 in	DEPTH 48 in	HEIGHT (RIDGE) 133½ in	ROOF SPAN 120 in
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SKILL LEVEL Intermediate	EST. BUILD TIME 2–3 weekends	ASSEMBLY STEPS 13 Steps
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01 Tools & Safety Equipment

SAFETY FIRST

Always wear eye protection, hearing protection, and a dust mask when cutting or drilling. Secure all workpieces with clamps before cutting. Never operate power tools while fatigued.

POWER TOOLS

Circular saw	× 1
Miter saw (recommended)	× 1
Jigsaw (for panel notches)	× 1
Impact driver	× 1
Power drill	× 1
Brad nailer (optional)	× 1

HAND TOOLS & LAYOUT

25 ft tape measure	× 1
Speed square / framing square	× 1
Combination square	× 1
Pencil / lumber crayon	× 2
Chalk line	× 1
Level (4 ft minimum)	× 1

CLAMPING & WORKHOLDING

Bar clamps (24"–36")	× 4
F-clamps	× 4
Sawhorses / work table	× 2
Roller stand (panel support)	× 1

SAFETY EQUIPMENT (REQUIRED)

Safety glasses / goggles	× 1
Ear protection (earmuffs)	× 1
N95 / P100 dust mask	× pack
Work gloves	× 1 pair
Steel-toe boots	× 1 pair

02 Hardware & Fasteners List

ITEM	SPECIFICATION	QTY (EST.)	NOTES
Wood screws — structural	3" #10 Exterior deck screws	100 pcs	Post/beam connections
Wood screws — framing	2½" #9 Exterior screws	400 pcs	Shelf frames, rafters, trim

ITEM	SPECIFICATION	QTY (EST.)	NOTES
Wood screws — siding	1½" #8 Exterior screws	500 pcs	Panel attachment
Heavy-duty door hinges	4" heavy butt hinges, galv.	8 pcs (4/door)	2 doors × 4 hinges each
Door handles / pulls	T-bar or barn pull, 6"	2 pcs	One per door
Door latch / hasp	Barrel bolt or padlock hasp	1 set	Center door closure
Roofing screws (metal)	1½" self-tapping hex head	100 pcs	Corrugated metal sheets
Structural corner brackets	Simpson or equiv., galv.	8 pcs	Post-to-beam joints (optional but recommended)
Joist hanger (2×6)	LUS26 or equiv.	4 pcs	Joists A & B to beams
Exterior wood glue	Titebond III or equiv.	1 bottle	Panel lamination joints
Exterior primer & paint	Acrylic exterior, colour TBD	2 gallons	All exposed wood surfaces
Corrugated metal sheet	4'×8' galvanized steel	3 sheets	Roofing step 12

03

Complete Cut List

All dimensions given as **Length × Width × Thickness** in inches. Nominal lumber dimensions shown in the Material column. Actual finished dimensions assumed for all calculations.

CUT LIST A — STRUCTURAL FRAMING

MARK	PART NAME	QTY	LENGTH	WIDTH	THICK	MATERIAL	NOTES
A	4×4 Post A	2	120"	3½"	3½"	4×4 lumber	Front tall posts

MARK	PART NAME	QTY	LENGTH	WIDTH	THICK	MATERIAL	NOTES
B	4×4 Post B	2	108"	3½"	3½"	4×4 lumber	Rear shorter posts
C	4×4 Beam C	2	96"	3½"	3½"	4×4 lumber	Top cross-beams
A	Joist A	2	89"	5½"	1½"	2×6 lumber	Long floor joists
B	Joist B	2	41"	5½"	1½"	2×6 lumber	Short floor joists
C	Brace Support C	4	14⅞"	3½"	1½"	2×4 lumber	45° cut both ends
A	Shelf Frame A	2	38"	3½"	1½"	2×4 lumber	Mid-shelf cross members
B	Shelf Frame B	2	96"	3½"	1½"	2×4 lumber	Mid-shelf long rails
A	Top Shelf Frame A	2	19¼"	3½"	1½"	2×4 lumber	Upper shelf cross members
B	Top Shelf Frame B	2	96"	3½"	1½"	2×4 lumber	Upper shelf long rails
C	Top Shelf Brace C	2	29⅜"	3½"	1½"	2×4 lumber	45° cut both ends
A	Roof Rafter A	7	85½"	3½"	1½"	2×4 lumber	Spaced 16–15/16" c/c
B	Roof Rafter B	2	85½"	3½"	1½"	2×4 lumber	Edge rafters (border)
B	Roof Face Board B	2	120"	3½"	1½"	2×4 lumber	Front & back fascia

✂ CUT LIST B — SHELF PANELS (3/4" PLYWOOD)

MARK	PART NAME	QTY	LENGTH	WIDTH	THICK	MATERIAL	NOTES
A	Bottom Shelf Panel A	1	96"	48"	¾"	¾" Plywood	Notch corners for posts — see diagram
B	Middle Shelf Panel B	1	96"	41"	¾"	¾" Plywood	No notching required
C	Top Shelf	1	96"	22¾"	¾"	¾"	No notching required

MARK	PART NAME	QTY	LENGTH	WIDTH	THICK	MATERIAL	NOTES
	Panel C					Plywood	

Ø CUT LIST C – WALL SIDING & TRIM (3/4" PLYWOOD)

MARK	PART NAME	QTY	LENGTH	WIDTH	THICK	MATERIAL	NOTES
A	L/R Side Panel A	38	48"	5"	¾"	¾" Plywood	Horizontal siding boards
B	L/R Side Panel B	2	48"	5"	¾"	¾" Plywood	Fill-in siding at top angle
C	L/R Side Panel C	2	32 ³ / ₁₆ "	5"	¾"	¾" Plywood	Angle-cut top piece per side
A	Back Siding Panel A	18	97½"	5"	¾"	¾" Plywood	Horizontal — requires 10' sheets
A	Door Trims A	2	105"	3½"	¾"	¾" Plywood	Vertical door surround (sides)
B	Door Trims B	1	90½"	3½"	¾"	¾" Plywood	Horizontal door surround (top)
A	Inner Trim A	1	89"	3½"	1½"	2×4 lumber	Horizontal inside header
B	Inner Trim B	2	93¾"	3½"	1½"	2×4 lumber	Vertical inside door jambs

Ø CUT LIST D – DOOR ASSEMBLIES (3/4" PLYWOOD + 2×4)

MARK	PART NAME	QTY	LENGTH	WIDTH	THICK	MATERIAL	NOTES
A	Door Panel A	4	101½"	4⅝"	¾"	¾" Plywood	Narrow edge boards, 2/door
B	Door Panel B	12	101½"	6"	¾"	¾" Plywood	Main vertical boards, 6/door
A	Door Frame A	4	90 ¹¹ / ₁₆ "	3½"	1½"	2×4 lumber	Vertical door frame stiles, 2/door
B	Door Frame B	4	42 ¹⁵ / ₁₆ "	3½"	1½"	2×4 lumber	Horizontal door rails, 2/door

PANEL SPACING NOTE – DOORS

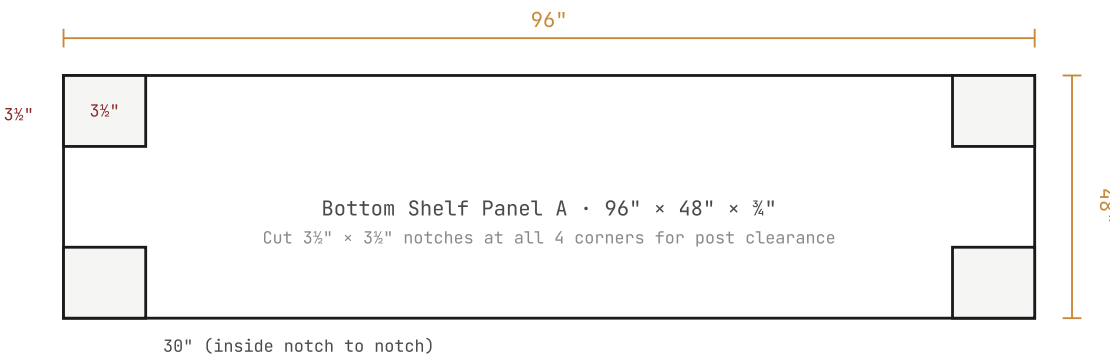
Bottom gap from door frame: 6¼" · Side inset from frame edge: 2-5/16" · Top gap: 1-9/16". Door boards are placed vertically with panels flush to the front face of the door frame.

04 Critical Technical Guidelines for Cutting

READ BEFORE CUTTING ANYTHING

Verify all measurements against the original plan before each cut. Always measure twice, cut once. Label every piece immediately after cutting to avoid confusion during assembly.

0 BOTTOM SHELF PANEL A – POST NOTCH DETAIL



Corner notch reference: 10-11/16" from panel edge to notch inner face (horizontal). 9-3/8" from bottom edge. Use jigsaw for clean corner cuts.

0 ANGLE CUTS – BRACE SUPPORT C & TOP SHELF BRACE C

BRACE SUPPORT C – 14⅜" @ 45°

Set miter saw to exactly 45°. Cut both ends in the same direction (parallel miter). Verify the finished length is 14-3/8" measured from long point to long point. Repeat for all 4 pieces.

TIP

Gang all 4 braces together and make one cut for consistent angles.

TOP SHELF BRACE C – 29⅜" @ 45°

Same procedure as brace support C. Set miter to 45°. Cut both ends parallel. Finished length: 29-3/8" long-point to long-point. Verify flat against top shelf frame before installing.

WARNING

Do not confuse with Brace Support C — different lengths.

3" #10	4×4 Post A/B to 4×4 Beam C — structural connection, pre-drill to avoid splitting	3 screws / joint
2½" #9	Joist A/B to Beams — toe-screw from both sides at 45° angle	2 screws / end
2½" #9	Shelf frame A to Shelf frame B — butt joint, flush to outside edge	3 screws / joint
2½" #9	Brace Support C corners — angle-cut into joist/frame corner	2 screws / end
2½" #9	Roof rafter A/B to face boards — pre-drill, 2 screws per rafter end	2 screws / end
1⅝" #8	Siding panels to framing — 2 screws per stud crossing, stagger pattern	2 screws / stud
1⅝" #8	Door panels to door frame — 2 screws per panel per rail	2 screws / panel end
2½" #9	Inner trim A/B to framing — face screw through trim into stud	3 screws / stud
2½" #9	Door trim A/B (face trim) to framing — keep flush to siding face	3 screws / stud

PRE-DRILLING RULE

Pre-drill all connections within 2" of any board end using a bit 1/64" smaller than the screw shank diameter. This prevents splitting, especially in 4×4 posts. Countersink all exterior face screws 1/8" below surface and fill with exterior wood filler before painting.

05 Step-by-Step Assembly

BEFORE YOU BEGIN

Have all pieces cut and labelled. Work on a flat, level surface. Dry-fit each sub-assembly before driving any fasteners. Apply exterior primer to all cut ends before assembly.

1

Post & Beam Frame

Stand 4×4 Post A (120") and Post B (108") in pairs for front and back frames. Attach 4×4 Beam C (96") across the top of each pair — front beam connects the two Post A tops, rear beam connects the two Post B tops. Posts B are shorter to create the roof slope. Use 3" wood screws, 3 per joint. Pre-drill to prevent splitting. Brace diagonally with temporary 2×4 while fastening to maintain square.

3" screws

3 screws/joint

pre-drill required

check square

2

Bottom Support Frame (Joist Assembly)

Assemble the floor joist frame: Joist A (89") runs front-to-back; Joist B (41") runs side-to-side forming a rectangle. Install 4 Brace Support C pieces at each inside corner — both ends cut at 45°, installed diagonally for racking resistance. Connect to the post assembly at floor level. Use 2½" screws for brace ends; use joist hangers at beam connections for additional strength. Reference view: Joist B inside Joist A, braces in corner quadrants.

2½" screws

45° brace cuts

verify rectangle is square (diagonal measure)

3

Middle Shelf Support Frame

Build the mid-level shelf frame by connecting Shelf Frame A (38") as cross-members to Shelf Frame B (96") rails. Use 2½" wood screws, 3 per joint. Position the shelf frame on the post assembly: place the lower cross-member 18½" from the floor; the upper cross-member at 37" from the floor (18½" above the lower one). This creates the bottom and middle shelf positions. Fasten to posts with 2½" screws toe-screwed at 45°.

2½" screws

Lower shelf @ 18½" from floor

Upper shelf @ 37" from floor

4

Top Shelf Support Frame

Assemble the upper shelf frame using Top Shelf Frame A (19¼") as short cross-members and Top Shelf Frame B (96") as long rails. Add Top Shelf Brace C (29¾", 45° both ends) diagonally within the frame to prevent racking — one brace per end. Fasten with 2½" screws. Install the completed top frame assembly onto the posts at the appropriate height, maintaining the sloped-roof geometry. The brace angle cut should sit flush against both frame members.

2½" screws

45° brace cuts

verify brace sits flush before fastening

5

Bottom Shelf Panel Installation

Cut Bottom Shelf Panel A (96" × 48" × ¾") with 3½" × 3½" notches at all 4 corners to fit around the 4×4 posts. Notch horizontal offset: 10–11/16" from panel edge. Notch vertical offset: 9–3/8" from bottom. Use a jigsaw for clean inside corners. Drop panel into position

resting on the Joist A/B frame. Secure with 1½" screws every 8" along each joist. Ensure panel surface is flush and level.

1½" screws @ 8" o/c

jigsaw for notches

test-fit before fastening

6

Middle Shelf Panel

Cut Middle Shelf Panel B (96" × 41" × ¾"). No post notching required — this panel sits between the posts. Lay onto the lower shelf frame at the 18½" position. Fasten with 1½" screws every 8" into the shelf frame below. Check that panel sits flat and does not rock. Apply a thin bead of exterior wood glue along the frame rails before setting the panel for additional holding power.

1½" screws @ 8" o/c

exterior glue on rails

no notching needed

7

Top Shelf Panel

Cut Top Shelf Panel C (96" × 22¼" × ¾"). This narrower panel installs on the upper shelf frame against the rear of the structure. No post notching required. Fasten with 1½" screws every 8" into the Top Shelf Frame B below. The angled roof geometry means the front edge of this shelf aligns with the inner face of the Top Shelf Frame A cross-members. Verify alignment visually from the front before final fastening.

1½" screws @ 8" o/c

align front edge to frame

8

Roof Rafter Assembly & Installation

Build the roof rafter panel flat on the ground. Attach 2 Roof Face Board B (120") as the long front and rear fascias. Install 7 Roof Rafter A (85½") and 2 Roof Rafter B (85½") between them, spaced 16-15/16" center-to-center. The two Rafter B pieces go at the far left and right edges. Pre-drill rafter ends; use 2½" screws, 2 per rafter end. Once assembled, lift the roof onto the post tops. The roof projects 12" beyond the front posts. Attach with 3" screws through fascia into the tops of all posts.

2½" screws for assembly

3" screws to posts

16-15/16" rafter spacing c/c

2-person lift required

9

Wall Siding — Left, Right & Back

Left & Right Walls: Begin from the bottom. Install Side Panels A (48" × 5") horizontally with 1½" screws, 2 screws per post. Maintain a 1/8" reveal gap between each board for drainage. Continue up the wall. At the top, the angle cut pieces (Panel B flush, Panel C at the roof angle) close the wall. Cut Panel C to match the roof pitch angle at the top edge using a circular saw set to the appropriate bevel.

Back Wall: Install Back Siding Panel A (97½" × 5") horizontally from bottom to top, same reveal gap approach. These panels span the full 97½" width. Use 1½" screws into posts and beams.

1½" screws

1/8" gap between boards

measure roof pitch angle before cutting Panel C

10

Inner Trim Installation

Install Inner Trim A ($89" \times 3\frac{1}{2}" \times 1\frac{1}{2}"$) horizontally across the top of the door opening on the interior face — this acts as the inside header. Then install 2× Inner Trim B ($93\frac{3}{4}" \times 3\frac{1}{2}" \times 1\frac{1}{2}"$) vertically on either side of the opening as inside door jambs. Fasten with $2\frac{1}{2}"$ screws, 3 per stud crossing. Flush the trim pieces with the inside face of the framing. The inner trim frames the doorway opening from the inside and provides a solid hinge-backing surface.

$2\frac{1}{2}"$ screws

3 screws/stud

flush to inside face

11

Face Trim (Door Surround)

Install Door Trims A ($105" \times 3\frac{1}{2}" \times \frac{3}{4}"$) vertically on the left and right exterior sides of the door opening, flush with the front face of the siding. Then install Door Trim B ($90\frac{1}{2}" \times 3\frac{1}{2}" \times \frac{3}{4}"$) horizontally across the top. Mitre the top corners at 45° for a clean joint. Fasten with $2\frac{1}{2}"$ screws, pre-drilled. The trim should overlap the edge of the siding slightly, covering the raw panel edges. Fill screw holes and caulk seams before painting.

$2\frac{1}{2}"$ screws

45° miter at top corners

caulk all seams

12

Metal Roofing Sheets

Install 3 pieces of 4'×8' corrugated galvanized metal sheet panels over the roof rafter assembly. Start from the low side (rear) and work toward the high front, overlapping each sheet by one full corrugation ridge (approximately 2"). Use $1\frac{1}{2}"$ self-tapping hex head roofing screws with EPDM washers, driven through the crown of each corrugation, 2 screws per rafter crossing. Do not over-tighten — the EPDM washer should compress slightly, not crush. Overlap front edge beyond fascia by $1\frac{1}{2}"$ minimum for drip edge.

$1\frac{1}{2}"$ roofing screws

1 corrugation overlap

start from low side

EPDM washer — do not over-tighten

13

Door Assembly & Hanging (×2)

Build each door flat on the ground: Lay 2× Door Frame A ($90\text{--}11/16"$) as vertical stiles and 2× Door Frame B ($42\text{--}15/16"$) as top/bottom rails forming a rectangle. Fasten with $2\frac{1}{2}"$ screws, 3 per corner. Then attach door panels face-on: start with 2× Door Panel A ($4\text{--}5/8"$ wide) at the edges, then fill with 6× Door Panel B (6" wide) between them. Bottom gap: $6\frac{1}{4}"$. Side setback: $2\text{--}5/16"$. Top gap: $1\text{--}9/16"$. Use $1\frac{5}{8}"$ screws, 2 per panel end into each rail.

Hanging: Install 4 heavy-duty exterior door hinges per door on the inner trim jambs. Use shims to set a $1/8"$ reveal gap around each door. Mark and pre-drill all hinge holes before driving screws. Install door handles centred at approx. 42" from floor. Add barrel bolt or hasp at door meeting edge.

$2\frac{1}{2}"$ screws — frame

$1\frac{5}{8}"$ screws — panels

4 hinges / door

06

Material & Cost Estimate

Prices are approximate US retail averages. Actual costs will vary by region, supplier, and current market conditions. All lumber quantities include a 10% overage allowance for waste and miscuts.

NOTE ON LUMBER LENGTHS

Some siding pieces (Back Siding Panel A at 97½", Door Trims A at 105") exceed standard 8-foot (96") sheets. Source 4'×10' plywood sheets or buy 5'×10' and rip to width. Alternatively, butt-join shorter pieces with a backing block mid-span.

CATEGORY	ITEM / SIZE	UNIT	QTY NEEDED	UNIT PRICE (USD)	TOTAL (USD)
LUMBER	4×4 × 10 ft (structural posts/beams)	each	7	\$18.00	\$126.00
	2×6 × 8 ft (floor joists)	each	4	\$11.00	\$44.00
	2×4 × 8 ft (framing, rafters, trim, door frames)	each	32	\$6.50	\$208.00
	2×4 × 12 ft (long rails, face boards)	each	6	\$9.50	\$57.00
PLYWOOD	¾" CDX Plywood 4×8 (shelf panels, siding strips)	sheet	10	\$58.00	\$580.00
	¾" Plywood 4×10 (back siding, door trim, door panels)	sheet	6	\$75.00	\$450.00
ROOFING	Galvanized corrugated metal sheet 4'×8'	sheet	3	\$42.00	\$126.00
HARDWARE	3" exterior deck screws (box of 100)	box	2	\$14.00	\$28.00
	2½" exterior screws (box of 250)	box	3	\$16.00	\$48.00
	1½" exterior screws (box of 250)	box	3	\$13.00	\$39.00

CATEGORY	ITEM / SIZE	UNIT	QTY NEEDED	UNIT PRICE (USD)	TOTAL (USD)
	1½" roofing screws with EPDM washer (box of 100)	box	1	\$16.00	\$16.00
	Heavy-duty 4" exterior door hinges	each	8	\$8.50	\$68.00
	Barn-style door handles / pulls	each	2	\$14.00	\$28.00
	Door hasp & padlock (closure)	set	1	\$18.00	\$18.00
	Joist hangers LUS26 (2×6)	each	4	\$3.50	\$14.00
FINISH	Structural corner brackets (galvanized)	each	8	\$4.50	\$36.00
	Exterior wood primer	gallon	1	\$32.00	\$32.00
	Exterior latex paint (colour TBD)	gallon	2	\$42.00	\$84.00
	Exterior wood filler / caulk	tube	4	\$7.00	\$28.00
	Exterior wood glue (Titebond III)	bottle	1	\$18.00	\$18.00
SUBTOTAL — MATERIALS					\$2,048.00
Contingency / miscellaneous (10%)					~\$205.00

Estimated Total: ≈ \$2,250 USD

▢ LUMBER LINEAR FOOTAGE SUMMARY

4×4 LUMBER (TOTAL LINEAR FEET) 4×4 Post A (2 × 10 ft)20 lf 4×4 Post B (2 × 9 ft)18 lf 4×4 Beam C (2 × 8 ft)16 lf Total54 lf	2×6 LUMBER (TOTAL LINEAR FEET) Joist A (2 × 7.4 ft)14.8 lf Joist B (2 × 3.4 ft)6.8 lf Total21.6 lf
2×4 LUMBER KEY QUANTITIES (LF)	PLYWOOD SHEET COUNT (¾")

Roof rafters A + B (9 × 7.1 ft)	63.7 lf
Roof face boards B (2 × 10 ft)	20 lf
Door frames A+B (8 pcs)	37.6 lf
Shelf/trim/brace (all others)	~75 lf
Total 2×4	~196 lf

Shelf panels A, B, C	3 sheets
Left/right siding strips	4 sheets (4×8)
Back siding (97½" pieces)	4 sheets (4×10)
Doors + door trims	2 sheets (4×10)
Total	10 × 4×8 + 6 × 4×10

07 Finishing & Final Notes

PAINT & WEATHERPROOFING

Sand all plywood edges with 80-grit before priming. Apply 2 coats exterior primer to all raw plywood edges. Two finish coats of exterior latex, sanding with 120-grit between coats. Pay special attention to the roof fascia edges and door panel tops — these are most vulnerable to moisture ingress.

DOOR GAP TOLERANCE

Allow 1/8" reveal all around each door. Wood will swell seasonally; too tight a fit in dry conditions means stuck doors in wet weather. If doors bind after humidity, plane the hinge side slightly. Do not plane the latch side — adjust hinges instead.

LEVELLING THE STRUCTURE

After setting in final position, check all four legs for level. Shim short posts with pressure-treated shim material. For permanent installation, consider concrete piers or treated post bases to keep posts off ground contact.

ROOF PITCH & DRAINAGE

The roof has a built-in slope (Post A 120" vs Post B 108" = 12" drop over 48" depth, approximately 25% grade). Ensure the low end (rear) has clearance for water runoff away from the structure. Grade the ground away from the base if needed.

BUILD SEQUENCE SUMMARY

1. Post/beam

2. Floor joists

3. Mid shelf

4. Top shelf

5-7. Panels

8. Roof

9. Siding

10-11. Trim

12. Metal roof

13. Doors